Company Background

What is ByteForge IT Hardware? The company offers graphics cards, storage solutions, monitors and computer accessories. It also offers technical services such as laptop cleaning and PC diagnostics.

Database Management System In order to keep track of business details like customer information, employee records, product stock and sales transactions a database system is used by ByteForge . This system enables the company to classify data adequately, correct errors and make work more efficient.

Objectives

The main objectives of this system are to create database management systems as below:

- To manage customer, employee, product, and sales data in an organized way.
- To reduce errors in recording and updating information.
- To make data access faster and easier for employees.
- To improve inventory and sales management.

Administrator

- The administrator has full access to the system.
- Can view, add, edit, and delete all records in the database.
- Manages employee accounts and system security.
- Monitors overall system performance and data accuracy.

Manager

- The manager can view, add, edit, and delete product and sales records.
- Can monitor inventory and check stock levels.
- Can view employee records and sales reports.
- Helps in managing daily business operations.

Employee

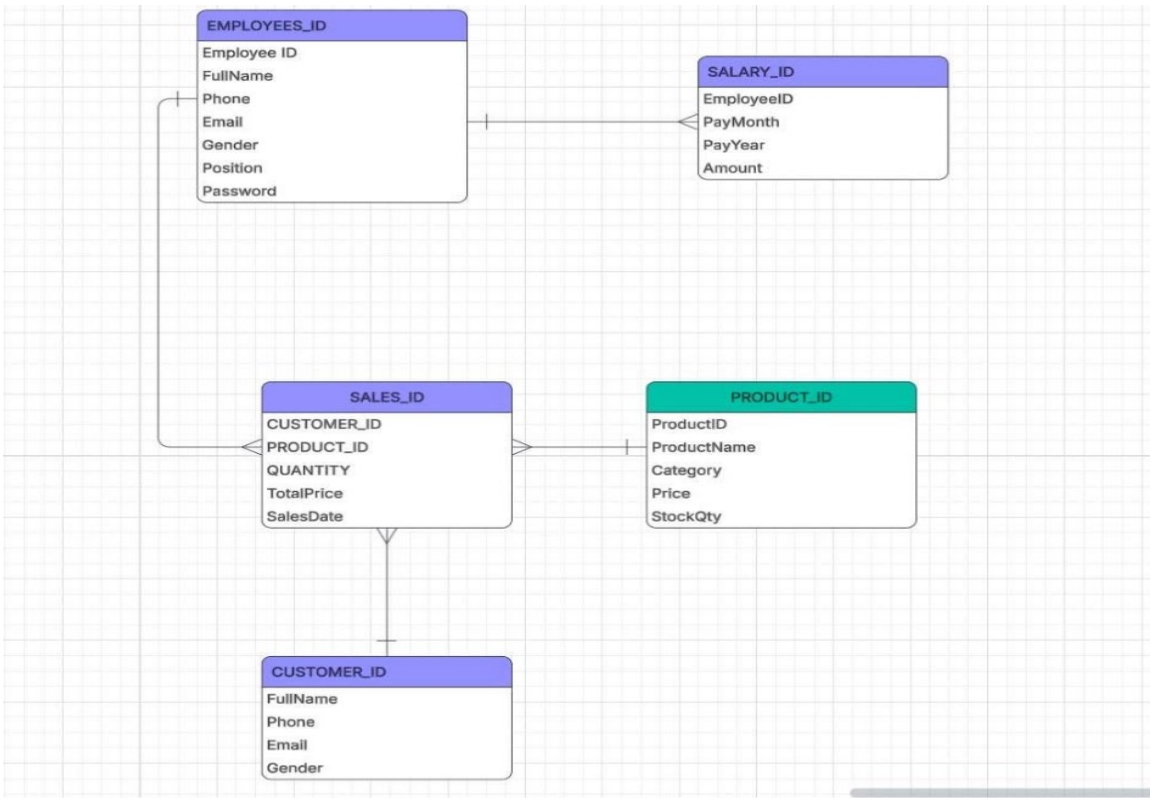
- Employees can add and manage customer records.
- Can record new sales transactions.
- Can view product information and stock levels.
- Can print sales reports when needed.

System Users

The ByteForge database application is built for three types of users: Administrator, Manager and Employee. There are various users in the system with different roles and activities. The system is totally under the control of the Administrator. The admin can control all of the data, users and ensure everything is in working order. Administrator is in charge of system security and data correctness. Matters concerning these men have been returned to the Manager for attention. The manager is able to control the products, check stock amount, sales and information about employees. The manager also utilizes the system to print sales and inventory statistics.

The Employee uses this system, in order to carry out his daily functions. Employee can add customer information, record sales, product details and print the sales report. The workforce member ensures that your business operations for the day are accurately posted in the system .

Entity Relationships Diagram



Data Dictionary

Sale Table

Attribute	Description	Data Type	Size	Data Format	Constraints
SaleID	Unique ID for each sale	VARCHAR2	7	S000	Primary Key
CustomerID	ID of the customer	VARCHAR2	7	C000	Foreign Key
ProductID	ID of the product	VARCHAR2	7	P000	Foreign Key
EmployeeID	Customer's purchse ID	VARCHAR2	7	E000	Foreign Key
Quantity	Number of products sold	NUMBER	3	Numeric Value	Not Null
SaleDate	Date of the sale	DATE	_	DD/MM/YYYY	Not Null
TotalPrice	Total price of the sale	CURRENCY	_	Currency (RM)	Not Null

Customer Table

Attribute	Description	Data Type	Size	Data Format	Constraints
CustomerID	Customer's purchase ID	VARCHAR2	7	C000	Primary Key
FullName	Customer full name	VARCHAR2	50	Characters	Not Null
Phone	Customer phone number	VARCHAR2	15	05XXXXXXXX	Unique
Email	Customer email address	VARCHAR2	50	name@email.com	Unique
Gender	Customer gender	VARCHAR2	10	Male / Female	Not Null

Product Table

Attribute	Description	Data Type	Size	Data Format	Constraints
ProductID	Unique ID for each product	VARCHAR2	7	P000	Primary Key
product_name	The product's name	VARCHAR2	50	Characters	Not Null
Category	Product category	VARCHAR2	50	Hardware/Accessory	Not Null
selling_price	The selling price of the products	CURRENCY	-	Currency (RM)	Not Null
StockQty	Quantity available in stock	NUMBER	4	Numeric Value	Not Null

Employee Table

Attribute	Description	Data Type	Size	Data Format	Constraints

EmployeeID	Unique ID for each employee	VARCHAR2	7	E000	Primary Key
FullName	Employee's name	VARCHAR2	50	Characters	Not Null
Position	Employee job position	VARCHAR2	30	Staff / Manager	Not Null
Phone	Employee phone number	VARCHAR2	15	05XXXXXXXX	Unique
Gender	Employee gender	VARCHAR2	10	Male / Female	Not Null
Password	Employee login password	VARCHAR2	20	Encrypted Text	Not Null

Salary Table

Attribute	Description	Data Type	Size	Data Format	Constraints
SalaryID	Unique ID for each salary record	VARCHAR2	10	SAL001E001	Primary Key
EmployeeID	ID of the employee	VARCHAR2	4	E001	Foreign Key
PayMonth	Month of salary payment	VARCHAR2	15	July / August	Not Null
PayYear	Year of salary payment	NUMBER	4	YYYY	Not Null
Amount	Salary amount paid to the employee	CURRENCY	-	Currency (RM)	Not Null

Tables

Sale Table

SaleID	CustomerID	ProductID	Quantity	TotalPrice	SaleDate	EmployeeID	Click to Add
S001	C001	P001	1	$1,600.00	11/1/2025	E001	
S002	C003	P004	2	$460.00	11/3/2025	E002	
S003	C002	P003	2	$500.00	11/5/2025	E003	
S004	C005	P002	1	$899.00	11/6/2025	E004	
S005	C004	P005	1	$310.00	11/8/2025	E005	
S006	C010	P006	1	$199.00	11/10/2025	E006	
S007	C019	P007	2	$318.00	11/11/2025	E007	
S008	C012	P008	1	$899.00	11/12/2025	E008	
S009	C017	P009	3	$87.00	11/13/2025	E009	
S010	C015	P010	1	$210.00	11/15/2025	E010	

Customer Table

CustomerID	FullName	Phone	Email	Gender	Click to Add
C001	Ahmad Faiz	017-2345678	faiz@gmail.com	Male	
C002	Nur Aisyah	013-5566778	aisyah@yahoo.com	Female	
C003	Lim Wei Jun	012-9988776	weijun@gmail.com	Male	
C004	Siti Zulaikha	011-7788992	zulaikha@gmail.com	Female	
C005	Rajesh Kumar	016-9887765	rajesh@outlook.com	Male	
C006	Farhan Hakim	017-1112233	farhan@gmail.com	Male	
C007	Melissa Tan	012-6667890	mel_tan@gmail.com	Female	
C008	Ameerul Ariff	019-9988771	ameerul@gmail.com	Male	
C009	Katherine Lee	014-8899001	kathlee@gmail.com	Female	
C010	Daniel Chong	013-8877665	danielc@yahoo.com	Male	
C011	Nur Balqis	018-3456712	balqis@gmail.com	Female	
C012	Hafiz Zain	012-1234569	hafizzain@mail.com	Male	
C013	Wong Mei Hui	017-5566123	meihui@yahoo.com	Female	
C014	Faizal Nordin	019-7001122	faizal@outlook.com	Male	
C015	Sarah Abdullah	011-5522334	sarahabd@gmail.com	Female	
C016	Aaron Lim	012-8899772	alimgaming@hotmail.com	Male	
C017	Puteri Sofea	013-7712345	psofea@gmail.com	Female	
C018	Muhammad Da	014-5637781	danish123@gmail.com	Male	
C019	Joanne Teh	016-8899456	joannet@gmail.com	Female	
C020	Harith Iskandar	017-6200987	harith.official@yahoo.com	Male	

Product Table

ProductID	ProductName	Category	Price	StockQty	Click to Add
P001	RTX 4060 GPU	Hardware	$599.00	12	
P002	Ryzen 5 5600X	Hardware	$899.00	8	
P003	16GB DDR4 RAM	Hardware	$250.00	20	
P004	512GB NVMe SSD	Hardware	$230.00	15	
P005	Power Supply 650W	Hardware	$310.00	10	
P006	Mechanical Keyboard	Accessories	$150.00	25	
P007	Gaming Mouse	Accessories	$120.00	30	
P008	27"" IPS Monitor	Accessories	$780.00	7	
P009	External 2TB HDD	Accessories	$350.00	14	
P010	Laptop Cleaning Service	Services	$50.00	999	
P011	Thermal Paste Replacement	Services	$40.00	999	
P012	Full PC Diagnostic Check	Services	$60.00	999	
p22	safas	Hardware	$5,224.00	4242	
r444	yhhhh	Hardware	$24,242.00	5454	

Employee Table

EmployeeID	FullName	Position	Phone	Gender	Password	Click to Add
E001	Aiman Razak	Sales Executive	017-9988776	Male	Aiman123	
E002	Nur Izzati	Technician	013-7788990	Female	Izzati456	
E003	Jason Lim	Technician	012-5656677	Male	Jason789	
E004	Aisyah Rahmar	Admin Clerk	014-9001122	Female	Aisyah2025	
E005	Daniel Wong	Sales Staff	017-6623344	Male	Wong5522	
E006	Sharmila Devi	Technician	016-5566772	Female	Devi8800	
E007	Haziq Ismail	Inventory Man	018-8899221	Male	Haziq007	
E008	Christine Tan	Customer Supp	019-3302211	Female	Chris990	
E009	Muhd Firdaus	Repair Speciali	017-2224456	Male	Firdaus55	
E010	Nadia Syafiqah	Sales Trainee	013-4455661	Female	Nadia321	
sgre						

Salary Table

SalaryID	EmployeeID	PayMonth	PayYear	Amount	Click to Add
SAL001E001	E001	July	2025	$2,500.00	
SAL001E002	E002	July	2025	$2,400.00	
SAL001E003	E003	July	2025	$2,300.00	
SAL001E004	E004	July	2025	$2,600.00	
SAL001E005	E005	July	2025	$2,200.00	
SAL001E006	E006	July	2025	$2,100.00	
SAL001E007	E007	July	2025	$2,700.00	
SAL001E008	E008	July	2025	$2,000.00	
SAL001E009	E009	July	2025	$1,950.00	
SAL001E010	E010	July	2025	$2,050.00	
SAL002E001	E001	August	2025	$2,550.00	
SAL002E002	E002	August	2025	$2,450.00	
SAL002E003	E003	August	2025	$2,350.00	
SAL002E004	E004	August	2025	$2,650.00	
SAL002E005	E005	August	2025	$2,250.00	
SAL002E006	E006	August	2025	$2,150.00	
SAL002E007	E007	August	2025	$2,750.00	
SAL002E008	E008	August	2025	$2,050.00	
SAL002E009	E009	August	2025	$2,000.00	
SAL002E010	E010	August	2025	$2,100.00	
SAL003E001	E001	September	2025	$2,600.00	
SAL003E002	E002	September	2025	$2,500.00	
SAL003E003	E003	September	2025	$2,400.00	
SAL003E004	E004	September	2025	$2,700.00	
SAL003E005	E005	September	2025	$2,300.00	
SAL003E006	E006	September	2025	$2,200.00	
SAL003E007	E007	September	2025	$2,800.00	
SAL003E008	E008	September	2025	$2,100.00	
SAL003E009	E009	September	2025	$2,050.00	
SAL003E010	E010	September	2025	$2,150.00	
SAL004E001	E001	October	2025	$2,650.00	

SAL004E002	E002	October	2025	$2,550.00	
SAL004E003	E003	October	2025	$2,450.00	
SAL004E004	E004	October	2025	$2,750.00	
SAL004E005	E005	October	2025	$2,350.00	
SAL004E006	E006	October	2025	$2,250.00	
SAL004E007	E007	October	2025	$2,850.00	
SAL004E008	E008	October	2025	$2,150.00	
SAL004E009	E009	October	2025	$2,100.00	
SAL004E010	E010	October	2025	$2,200.00	
SAL005E001	E001	November	2025	$2,700.00	
SAL005E002	E002	November	2025	$2,600.00	
SAL005E003	E003	November	2025	$2,500.00	
SAL005E004	E004	November	2025	$2,800.00	
SAL005E005	E005	November	2025	$2,400.00	
SAL005E006	E006	November	2025	$2,300.00	
SAL005E007	E007	November	2025	$2,900.00	
SAL005E008	E008	November	2025	$2,200.00	
SAL005E009	E009	November	2025	$2,150.00	
SAL005E010	E010	November	2025	$2,250.00	
SAL006E001	E001	December	2025	$2,750.00	
SAL006E002	E002	December	2025	$2,650.00	
SAL006E003	E003	December	2025	$2,550.00	
SAL006E004	E004	December	2025	$2,850.00	
SAL006E005	E005	December	2025	$2,450.00	
SAL006E006	E006	December	2025	$2,350.00	
SAL006E007	E007	December	2025	$2,950.00	
SAL006E008	E008	December	2025	$2,250.00	
SAL006E009	E009	December	2025	$2,200.00	
SAL006E010	E010	December	2025	$2,300.00	

Sample Queries

1. Detailed Sales Report

```
SELECT
   Sales.SaleID,
   Customers.FullName AS CustomerName,
   Products.ProductName,
   Products.Category,
   Sales.Quantity,
   Products.Price,
   (Sales.Quantity * Products.Price) AS CalculatedTotal,
   Sales.SaleDate
FROM
   (
      Sales
      INNER JOIN Customers ON Sales.CustomerID = Customers.CustomerID
   )
   INNER JOIN Products ON Sales.ProductID = Products.ProductID;
```

2. Employees Who Earn Above Average Salary

```
SELECT
   Employees.EmployeeID,
   Employees.FullName,
   Salary.Amount
FROM
   Employees
   INNER JOIN Salary ON Employees.EmployeeID = Salary.EmployeeID
WHERE
   (
      (
         (Salary.Amount) > (
            SELECT
               AVG(Amount)
            FROM
               Salary
         )
      )
   );
```

3. Salary Per Month

```
SELECT
   *
FROM
   Salary
WHERE
   Month = [Enter Month Name:];
```

4. Customers Who Bought More Than 2 Products

```
SELECT
   Customers.FullName,
   SUM(Sales.Quantity) AS TotalItemsBought
FROM
   Customers
   INNER JOIN Sales ON Customers.CustomerID = Sales.CustomerID
GROUP BY
   Customers.FullName
HAVING
   SUM(Sales.Quantity) > 2;
```

Creating Table

```
CREATE TABLE Customers (
   CustomerID VARCHAR2(5) PRIMARY KEY,
   FullName   VARCHAR2(100),
   Phone      VARCHAR2(20),
   Email      VARCHAR2(100),
   Gender     VARCHAR2(10)
);

CREATE TABLE Employees (
   EmployeeID VARCHAR2(5) PRIMARY KEY,
   FullName   VARCHAR2(100),
   Position   VARCHAR2(50),
   Phone      VARCHAR2(20),
   Gender     VARCHAR2(10),
   Password   VARCHAR2(50)
);
```

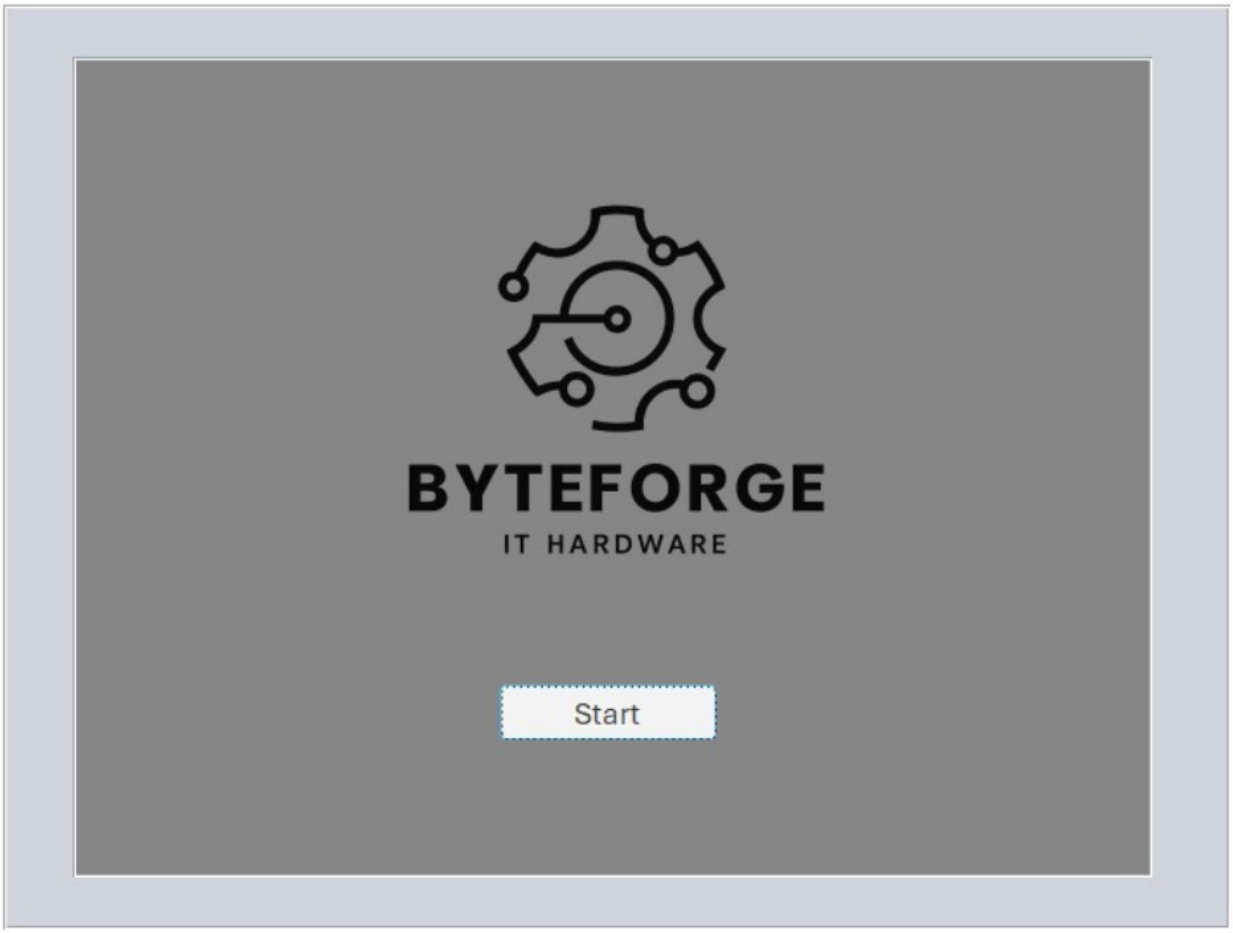
```
CREATE TABLE Products (
   ProductID   VARCHAR2(5) PRIMARY KEY,
   ProductName VARCHAR2(100),
   Category    VARCHAR2(50),
   Price       NUMBER(8,2),
   StockQty    NUMBER
);


CREATE TABLE Sales (
   SaleID     VARCHAR2(5) PRIMARY KEY,
   CustomerID VARCHAR2(5),
   ProductID  VARCHAR2(5),
   Quantity   NUMBER,
   TotalPrice NUMBER(10,2),
   SaleDate   DATE,
   CONSTRAINT fk_sales_customer FOREIGN KEY (CustomerID)
      REFERENCES Customers(CustomerID),
   CONSTRAINT fk_sales_product FOREIGN KEY (ProductID)
      REFERENCES Products(ProductID)
);

CREATE TABLE Salary (
   SalaryID   VARCHAR2(10) PRIMARY KEY,
   EmployeeID VARCHAR2(5),
   SalaryMonth VARCHAR2(15),
   Amount     NUMBER(10,2),
   CONSTRAINT fk_salary_employee FOREIGN KEY (EmployeeID)
      REFERENCES Employees(EmployeeID)
);
```

Display Data in Table

```
SELECT *
FROM customers;
SELECT *
FROM Employee;
SELECT *
FROM Products;
SELECT *
 FROM Salary;
SELECT *
FROM Sales;
```

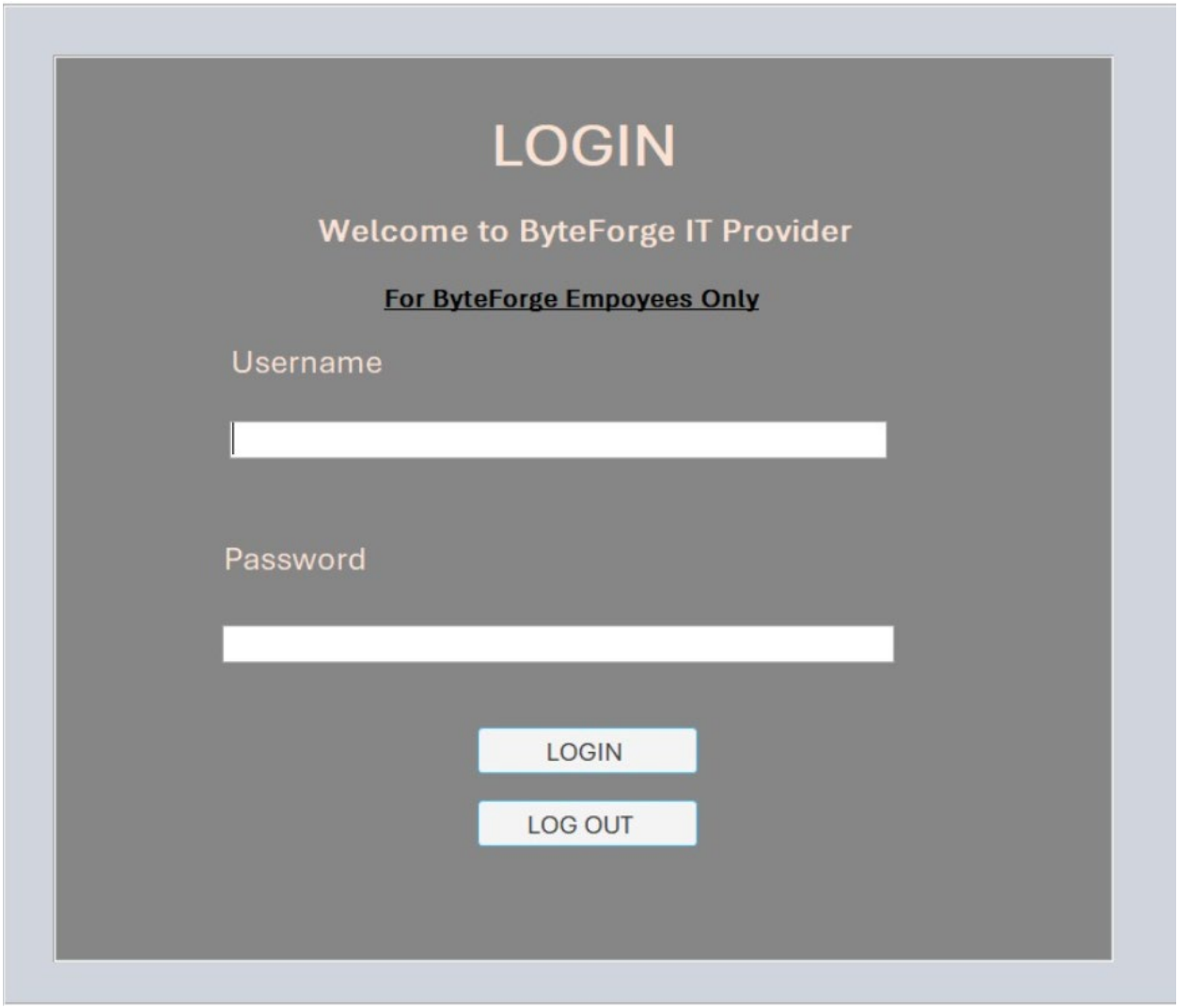
System Interface

Start Page



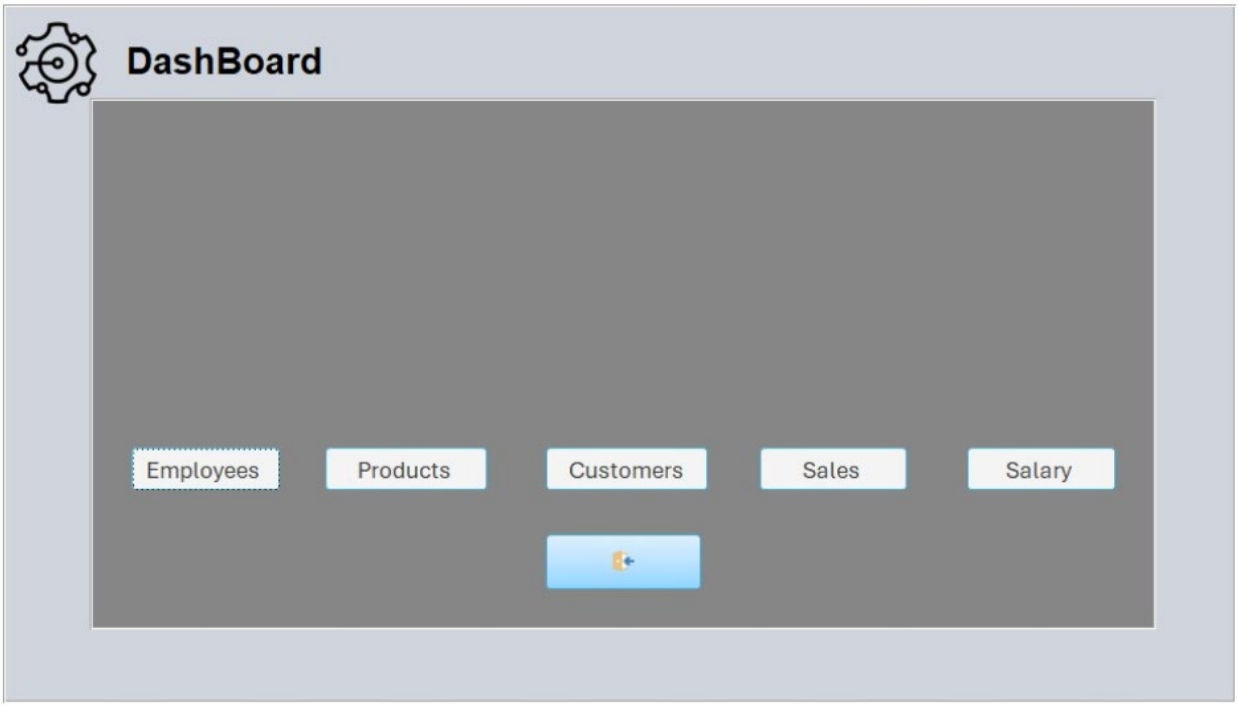
1. The Start Page is the first page displayed when the system is opened.
2. It shows the **ByteForge IT Hardware** logo.
3. The page contains a **Start** button.
4. When the user clicks the **Start** button, the system navigates to the **Login Page**.

Login Page



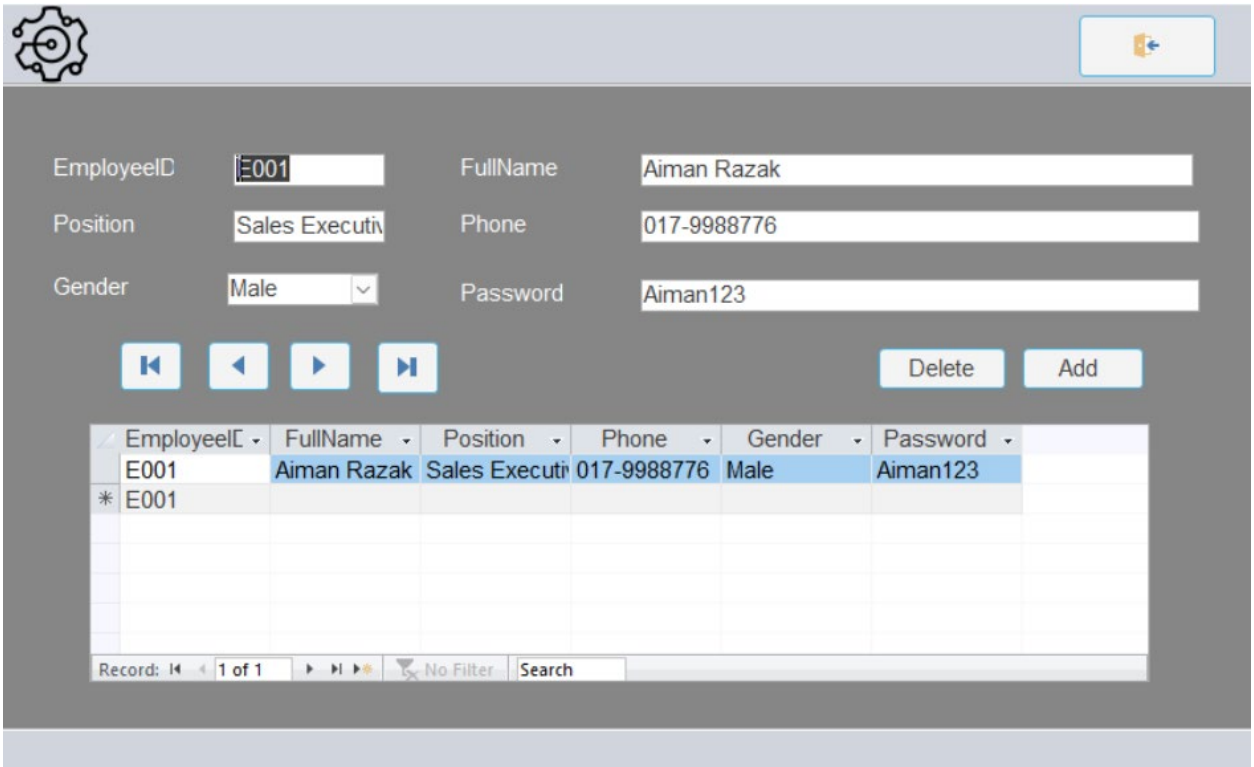
1. The Login Page is displayed after clicking the **Start** button.
2. This page is for **ByteForge employees only**.
3. Users must enter their **Username** and **Password** to access the system.
4. Clicking the **Login** button allows the user to enter the system if the details are correct.
5. Clicking the **Log Out** button exits the system

DashBoard



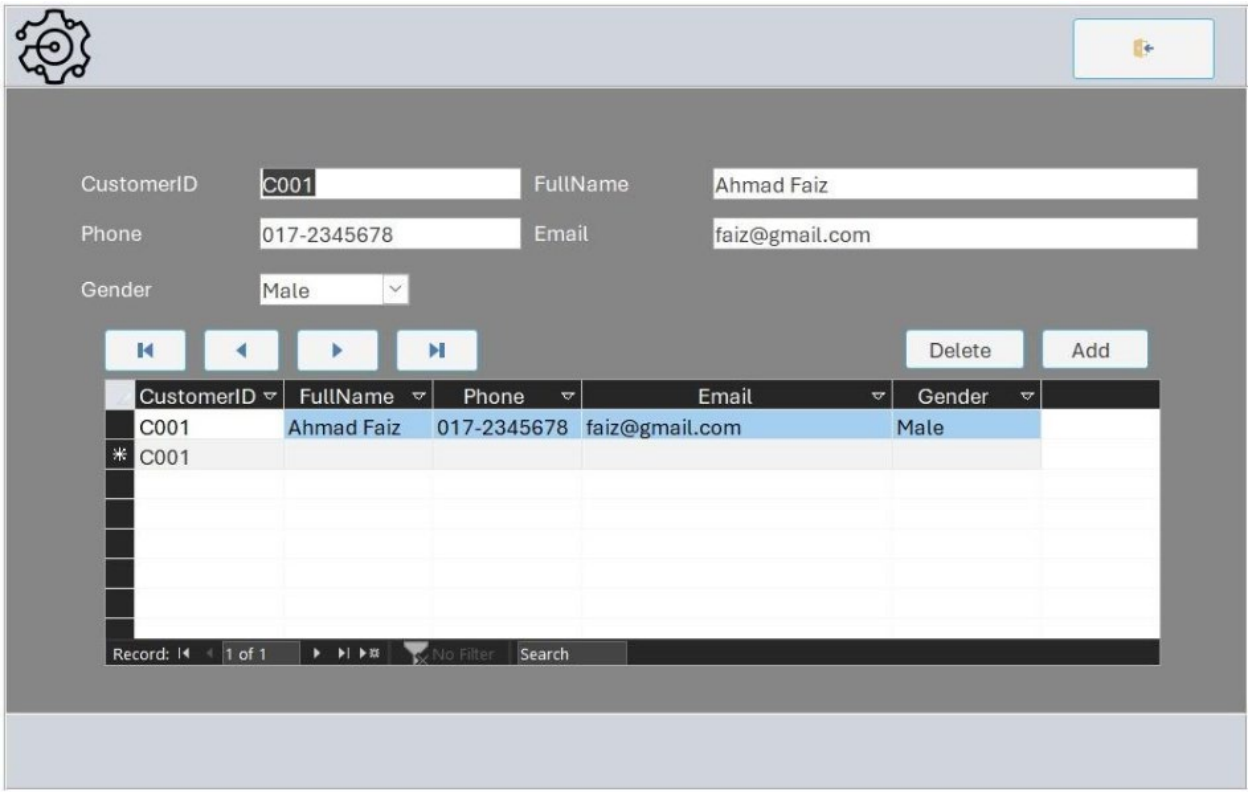
The DashBoard is the main navigation page of the system. It allows users to access different functions through buttons.
The page provides buttons for **Employees**, **Products**, **Customers**, Salary, and **Sales**, which direct the user to the selected section.

Employee Record Page



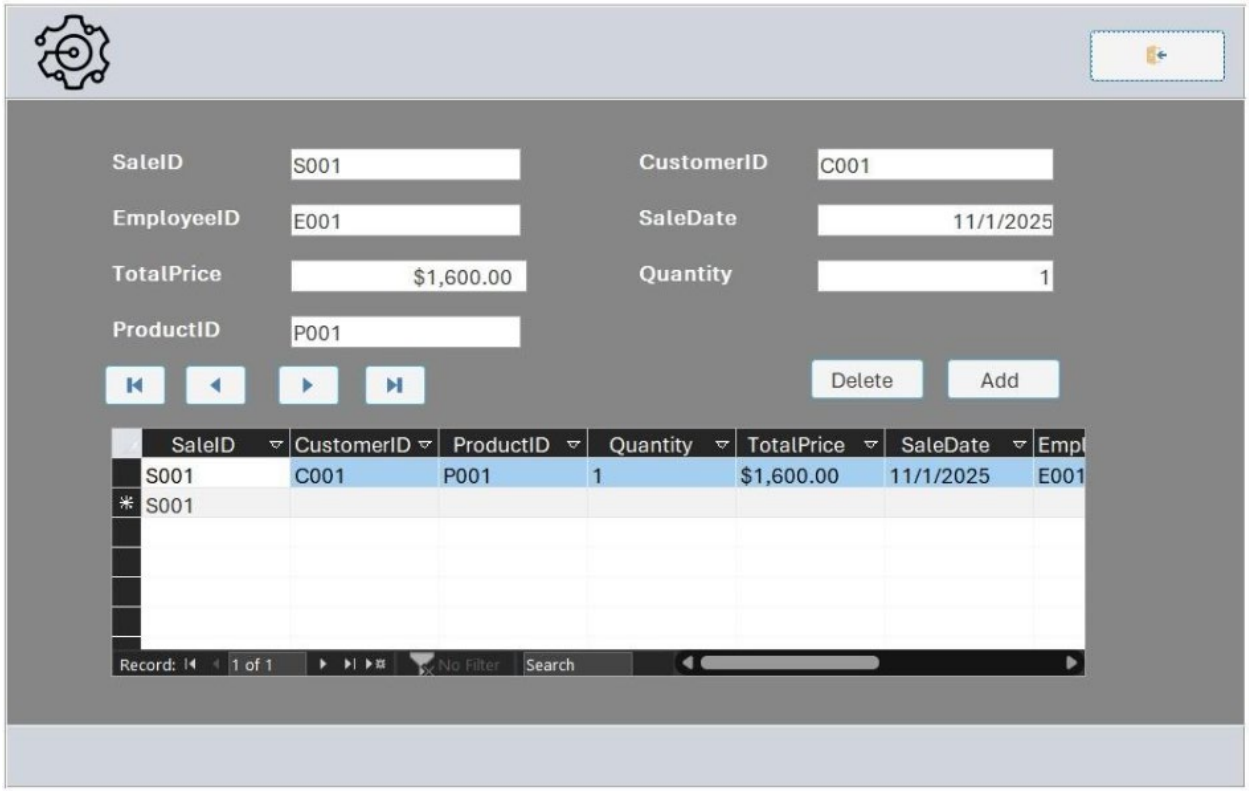
1. To open this page, the manager clicks the **"Employees"** button from the main menu or dashboard.
2. On this page, the manager can view all employee records stored in the system.
3. The information shown includes **Employee ID, Full Name, Position, Phone Number, and Gender**.
4. The manager can move between employee records using the navigation buttons at the bottom of the form.
5. To add a new employee, the manager clicks the **Add** button, fills in the fields, and then saves the record.
6. To update an existing employee, the manager selects the record, edits the information, and saves the changes.
7. To delete an employee, the manager selects the record and clicks the **Delete** button.
8. All changes made on this page are automatically updated in the **Employee** table in the database.

Customers view page



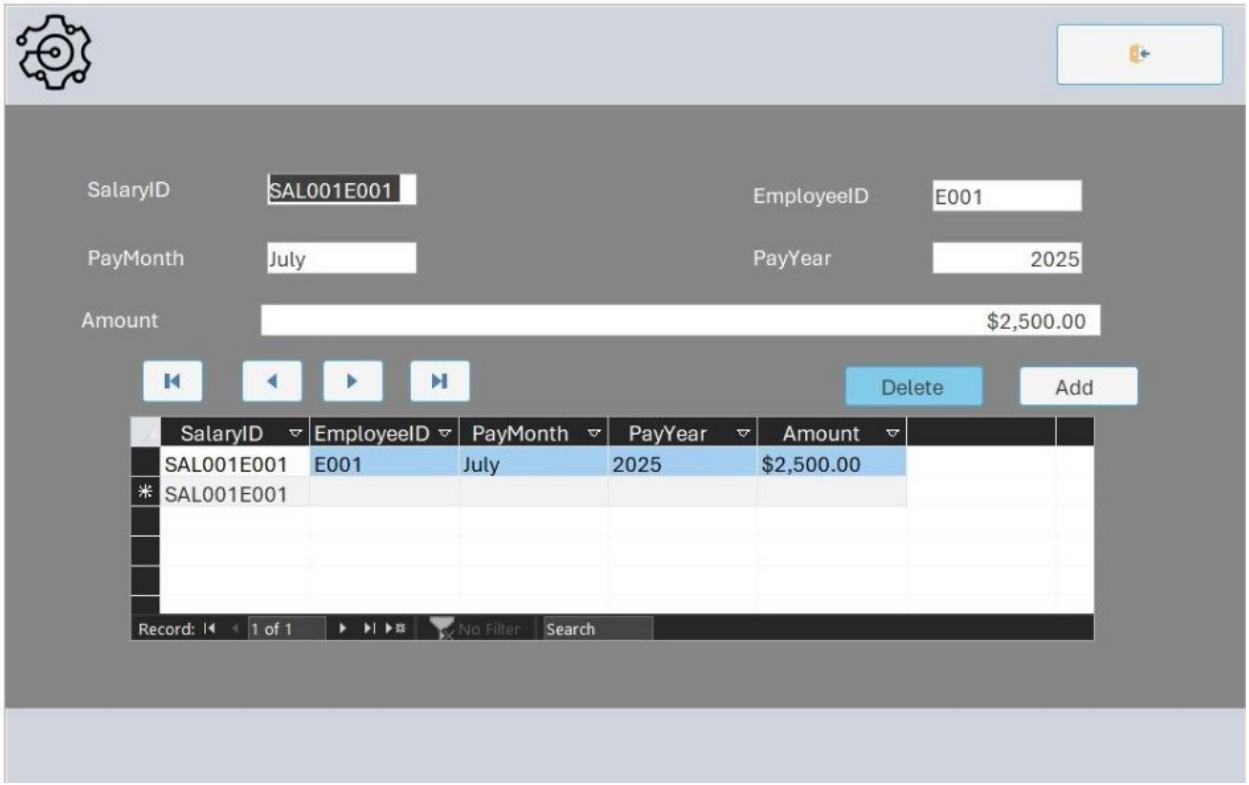
1-To open this page, the manager clicks the **"Customers"** button from the main menu or dashboard.
2-On this page, the manager can view all customer records stored in the system.
3-The information displayed includes **Customer ID, Full Name, Phone Number, Email, and Gender**.
4-The manager can navigate through the customer records using the navigation buttons provided on the form.
5-To add a new customer, the manager clicks the **Add** button, enters the customer details, and saves the record.
6-To update an existing customer, the manager selects the customer record, edits the information, and saves the changes.
7-To delete a customer, the manager selects the customer record and clicks the **Delete** button.
8-All changes made on this page are automatically updated in the **Customer table** in the database

Sales Page



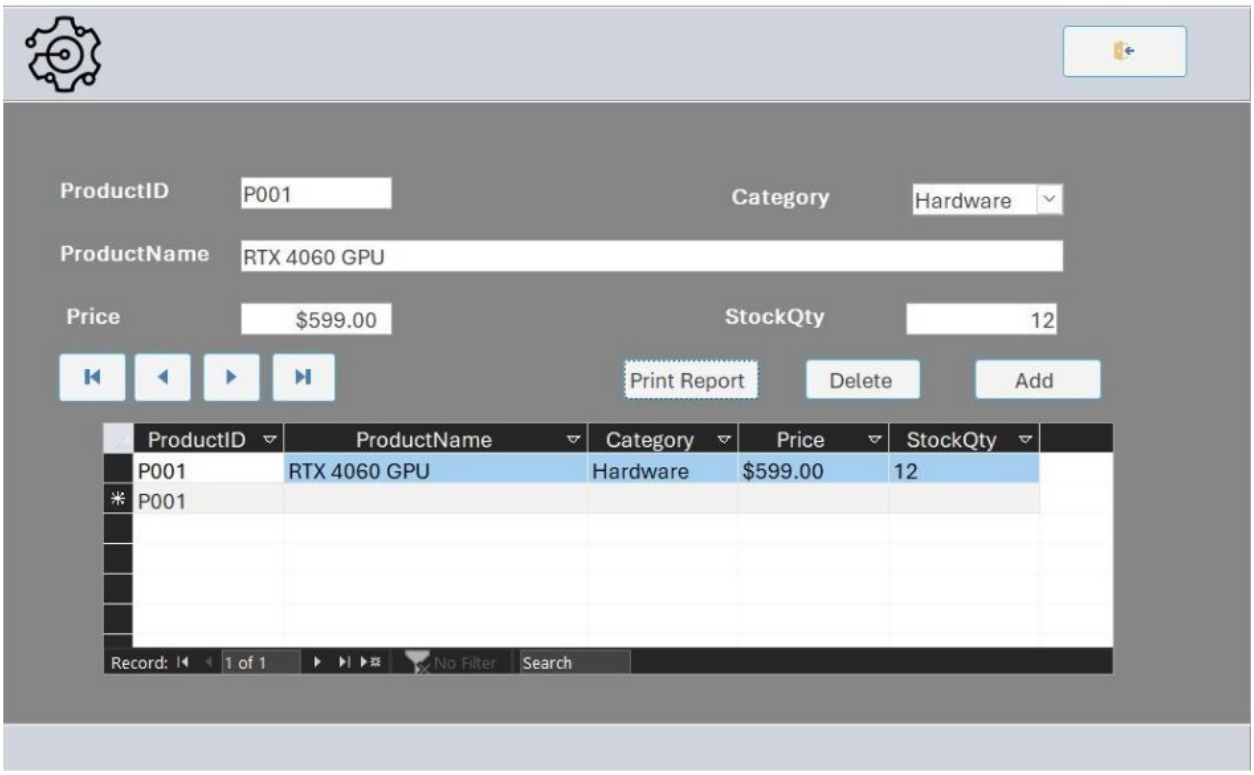
1-To open this page, the manager clicks the **"Sales"** button from the main menu or dashboard.
2-On this page, the manager can view all sales records stored in the system.
3-The information displayed includes **Sale ID, Customer ID, Product ID, Employee ID, Quantity, Total Price, and Sale Date**.
4-The manager can navigate through the sales records using the navigation buttons provided on the form.
5-To add a new sale record, the manager clicks the **Add** button, enters the required sale details, and saves the record.
6-To update an existing sale record, the manager selects the sale record, modifies the information, and saves the changes.
7-To delete a sale record, the manager selects the sale record and clicks the **Delete** button.
8-All changes made on this page are automatically updated in the **Sales table** in the database.

salary page



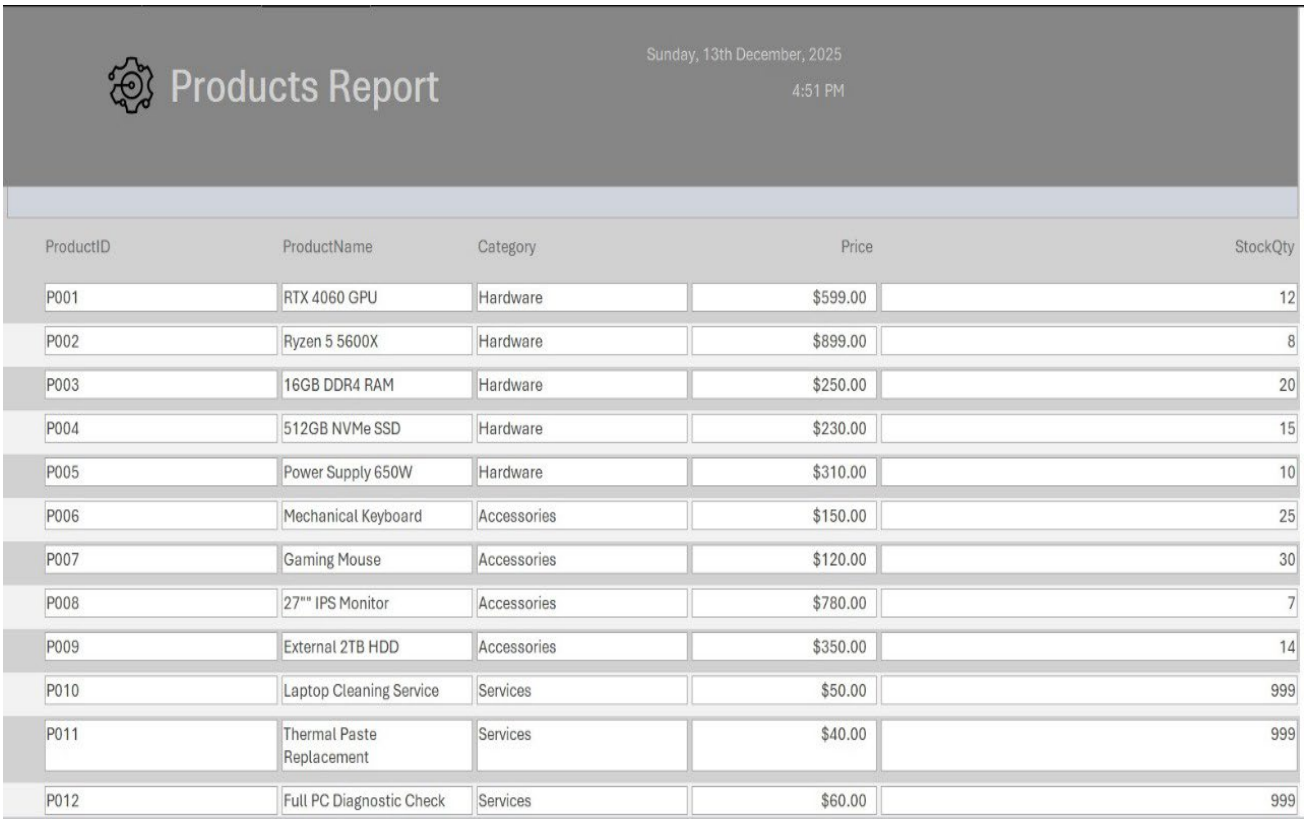
1-To open this page, the manager clicks the **"Salary"** button from the main menu or dashboard.
2-On this page, the manager can view all salary records stored in the system.
3-The information displayed includes **Salary ID, Employee ID, Pay Month, Pay Year, and Salary Amount**.
4-The manager can navigate through the salary records using the navigation buttons provided on the form.
5-To add a new salary record, the manager clicks the **Add** button, enters the required salary details, and saves the record.
6-To update an existing salary record, the manager selects the salary record, edits the information, and saves the changes.
7-To delete a salary record, the manager selects the salary record and clicks the **Delete** button.
8-All changes made on this page are automatically updated in the **Salary table** in the database.

Product view page



1-To open this page, the manager clicks the **"Products"** button from the main menu or dashboard.
2-On this page, the manager can view all product records stored in the system.
3-The information displayed includes **Product ID, Product Name, Category, Price, and Stock Quantity**.
4-The manager can navigate through the product records using the navigation buttons provided at the bottom of the form.
5-To add a new product, the manager clicks the **Add** button, enters the product details, and saves the record.
6-To update an existing product, the manager selects the product record, edits the information, and saves the changes.
7-To delete a product, the manager selects the product record and clicks the **Delete** button.
8-Additionally, the manager can generate a printed report of the product list by clicking the **Print Report** button.
9-All changes made on this page are automatically updated in the **Product table** in the database.

Products Report



Products Report

Sunday, 13th December, 2025

4:51 PM

ProductID	ProductName	Category	Price	StockQty
P001	RTX 4060 GPU	Hardware	$599.00	12
P002	Ryzen 5 5600X	Hardware	$899.00	8
P003	16GB DDR4 RAM	Hardware	$250.00	20
P004	512GB NVMe SSD	Hardware	$230.00	15
P005	Power Supply 650W	Hardware	$310.00	10
P006	Mechanical Keyboard	Accessories	$150.00	25
P007	Gaming Mouse	Accessories	$120.00	30
P008	27"" IPS Monitor	Accessories	$780.00	7
P009	External 2TB HDD	Accessories	$350.00	14
P010	Laptop Cleaning Service	Services	$50.00	999
P011	Thermal Paste Replacement	Services	$40.00	999
P012	Full PC Diagnostic Check	Services	$60.00	999

1. Employees can view all products available in the inventory.
2. The report displays product details such as Product ID, Product Name, Category, Price, and Stock Quantity.
3. Employees can print the product list by clicking the **"Print Products Report"** button.
4. To return to the previous page, click the **"Back to Inventory"** button.

Conclusion

The ByteForge IT Hardware Database Management System effectively offers a means for important business data to be managed in a proper and efficient manner. This has been achieved with the help of data integration from various tables, including those that contain customer, employee, product, and sales information. Overall, this means that data will be stored in a correct manner and retrieved quickly when it is needed. User access security has been ensured in this system with different levels of access for Administrator, Manager, and Employees.

In conclusion, this database enhances business daily activities in various ways, including decreasing mistakes, simplifying sales and inventory management, as well as offering a systematic way of accessing information. With components like management of records, sales management, inventory management, as well as generating reports, it sounds effective in running the activities of the company